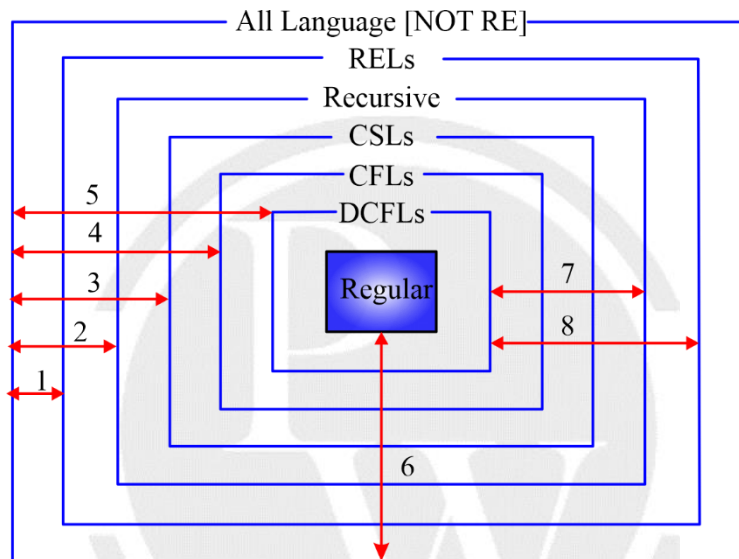


4

TURNING MACHINE

4.1 Classification of Languages



1. All languages which are not REs
2. All not recursive languages
3. All not CSLs
4. All not CFLs
5. All not DCFLs
6. All not regulars
7. All recursive languages which are not DCFLs
8. All REs which are not DCFLs

4.2 There are Two types of TM

- (a) DTM (Deterministic TM)
- (b) NTM (Non- Deterministic TM)

DTM

$$\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \begin{Bmatrix} L, R \\ \downarrow \quad \downarrow \\ \text{Left} \quad \text{Right} \end{Bmatrix}$$

NTM

$$\delta : Q \times \Gamma \rightarrow 2^{Q \times \Gamma \times \{L, R\}}$$

4.3 Unrestricted Grammar (UG) and Context Sensitive Grammar (CSG)

Unrestricted grammar (UG) also called as

- RE grammar

- Phase structure grammar

Context sensitive grammar (CSG) is also called as

- Non contracting grammar
- Bound restricted grammar

4.4 Grammars

Left Linear Grammar (LLG): $V \rightarrow VT^* \mid T^*$

Right Linear Grammar (RLG): $V \rightarrow T^*V \mid T^*$

Linear Grammar (LG): $V \rightarrow T^*VT^* \mid T^*$

Context Free Grammar (CFG): $LHS \rightarrow RHS, \quad |LHS| \leq |RHS|$

Unrestricted Grammar (UG): $LHS \rightarrow RHS$

4.5 Equivalence of various TMs

TM $\cong$ Single tape TM

TM $\cong$ One-way infinite tape TM

TM $\cong$ Two-way infinite tape TM

TM $\cong$ Multi tape and multi head TM

TM $\cong$ Universal TM

TM $\cong$ Two stack PDA

TM $\cong$ Multi stack PDA

TM $\cong$ FA with two stacks

TM $\cong$ FA + R/W tape + Bidirectional head

4.6 Restrictions on TM

(1) If TM tape is read only tape, this TM accepts regular.

TM $\cong$ FA (TM with read only tape)

(2) If TM head is unidirectional then $L(TM) = \text{Regular}$.

(3) If TM tape is read only and unidirectional head then $L(TM) = \text{Regular}$.

(4) If TM always halts then $L(TM)$ is recursive language.

(5) If TM always halts and uses linear bound tape then $L(TM)$ is CSL.

4.7 LBA, HTM and TM

HTM: It is a TM that always halts for every input.

TM: It halts for every valid string and for invalid strings either halts at “non final state” or “never halts”.

LBA: It is HTM but length of the tape we use linearly bounded.

- LBA accepts CSL languages.
- HTM accepts recursive languages.
- TM accepts Recursive Enumerable (RE) languages.

4.8 Recursively Enumerable Language

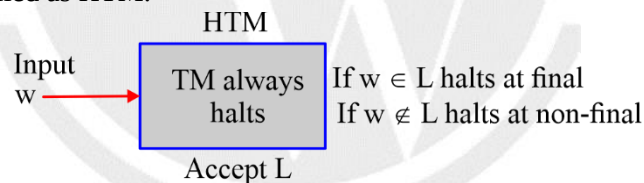
It is also called as:

- Enumerable language
- TM recognizable language
- TM Enumerable language
- Partially decidable language
- Semi-decidable language

4.9 Recursive Language

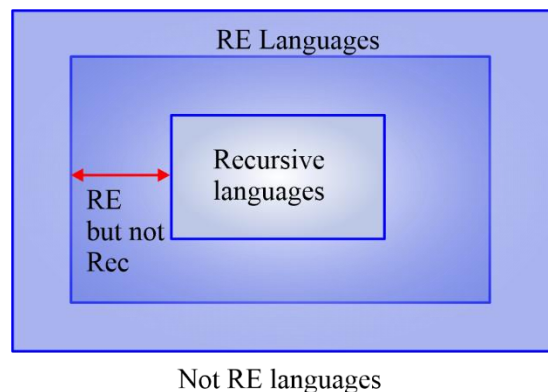
- Recursive language is acceptable by HTM, and hence acceptable by TM.
- Recursive also called as decidable language.
- Recursive also called as Turing decidable language.

If TM always halts, then TM is called as HTM.



4.10 Difference between Recursive and REL

- All Recursive languages are RE languages.



Note :

1. Union:

$$\text{REL} \cup \text{Finite} \Rightarrow \text{REL}$$

$$\text{REL} \cup \text{Regular} \Rightarrow \text{REL}$$

$$\text{REL} \cup \text{CFL} \Rightarrow \text{REL}$$

$$\text{REL} \cup \text{Recursive} \Rightarrow \text{REL}$$

$$\text{REL}_1 \cup \text{REL}_2 \Rightarrow \text{REL}$$

2. Intersection:

$$\text{REL} \cap \text{Finite} \Rightarrow \text{REL (Finite)}$$

$$\text{REL} \cap \text{CFL} \Rightarrow \text{REL}$$

$$\text{REL} \cap \text{Rec} \Rightarrow \text{REL}$$

$$\text{REL}_1 \cap \text{REL}_2 \Rightarrow \text{REL}$$

4.11 Closure Properties of Recursive languages

I. The following operations are not closed for recursive languages:

- Subset
- Substitution
- Homomorphism
- Finite substitution
- Infinite union
- Infinite intersection
- Infinite concatenation
- Infinite difference
- Infinite substitution

II. The following operations are closed for recursive languages:

- Complement
- Difference
- Finite difference
- Infinite followed by $\cup, \cap, -, \bullet, \subseteq, f$
- Remember not closed operations

4.12 Complement of Recursive Vs Complement of REL

